

An $O(\log^{1/6} n)$ Bound for the KLS Constant

Zhao Song*

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Abstract

The Kannan–Lovász–Simonovits (KLS) conjecture asks whether the KLS constant ψ_n of isotropic log-concave probability measures on $\mathbb{R}^n$ is bounded independently of n . The best previous upper bound is $\psi_n \lesssim \log^{1/4} n$, due to Letwin [Let26]. We prove $\psi_n \lesssim \log^{1/6} n$ and obtain the Poincaré bound $C_P(\mu) \lesssim \log^{1/3} n$ for every isotropic log-concave probability measure μ .

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1 Introduction

The Kannan–Lovász–Simonovits (KLS) conjecture is one of the central open problems in asymptotic convex geometry and the analysis of log-concave measures. In geometric language, it predicts that, up to a universal factor, the most economical way to bisect a convex body is already detected by a hyperplane. In analytic language, it asks whether every log-concave probability measure has a dimension-free spectral gap after its natural affine normalization. These formulations are equivalent up to universal constants. Besides its intrinsic geometric content, KLS governs concentration of Lipschitz functions and the conductance and mixing time of random walks used to sample from convex bodies; this algorithmic motivation is already present in the original work of Kannan, Lovász, and Simonovits [KLS95]. We refer to [LV19, KL25b] for broader surveys.

A probability measure μ on $\mathbb{R}^n$ is log-concave if

$$\mu(sA + (1-s)B) \geq \mu(A)^s \mu(B)^{1-s}, \quad 0 \leq s \leq 1,$$

for compact sets $A, B \subseteq \mathbb{R}^n$. Equivalently, on the affine hull of its support, μ has a density of the form e^{-V} with V an extended-real-valued convex function (allowing $V = +\infty$ outside the support). We call μ isotropic when its barycenter is zero and its covariance is the identity:

$$\int x d\mu(x) = 0, \quad \int x \otimes x d\mu(x) = I_n.$$

Every full-dimensional log-concave measure with nonsingular covariance has an isotropic image, unique up to an orthogonal transformation.

The Poincaré constant $C_P(\mu)$ is the least C for which

$$\mathrm{Var}_\mu[f] \leq C \int |\nabla f|^2 d\mu \tag{1}$$

holds for every locally Lipschitz function f with finite Dirichlet energy, with the additional requirement that every such f belongs to $L^2(\mu)$. If no finite C has this property, set $C_P(\mu) := +\infty$. For the full-dimensional laws considered below, Lemma 3.12 justifies passage from compactly supported smooth tests to this test class. On the isoperimetric side, let

$$\mu^+(A) := \liminf_{\varepsilon \downarrow 0} \frac{\mu(A + \varepsilon B_2^n) - \mu(A)}{\varepsilon},$$

and define the Cheeger constant and its reciprocal by

$$h_\mu := \inf_{0 < \mu(A) < 1} \frac{\mu^+(A)}{\min\{\mu(A), 1 - \mu(A)\}}, \quad \psi_\mu := h_\mu^{-1}.$$

For full-dimensional log-concave measures, Cheeger’s inequality and the reverse Cheeger–Buser inequality compare the two quantities in both directions [Che70, Bus82, Led04, Mil09]. With an explicit numerical comparison on the reverse side [PM21],

$$\frac{1}{4} \leq \frac{\psi_\mu^2}{C_P(\mu)} \leq \pi. \tag{2}$$

Thus, if $\psi_n := \sup\{\psi_\mu : \mu \text{ is isotropic and log-concave on } \mathbb{R}^n\}$, then the KLS conjecture is precisely $\sup_{n \geq 1} \psi_n < \infty$. Equivalently, every isotropic log-concave measure should satisfy $C_P(\mu) \leq C$. In affine-covariant form, if Σ is the covariance of an arbitrary log-concave measure, the conjecture predicts $C_P(\mu) \leq C \|\Sigma\|_{\mathrm{op}}$. The covariance dependence is necessary: testing Eq. (1) against linear functions gives $C_P(\mu) \geq \|\Sigma\|_{\mathrm{op}}$.

Development of the general bound. The historical asymptotic bounds below are stated for sufficiently large n . The localization lemma of Kannan, Lovász, and Simonovits reduced many high-dimensional integral inequalities to one-dimensional needles and gave the original estimate $\psi_n \lesssim \sqrt{n}$ [KLS95]. Bobkov connected isoperimetry to radial variance [Bob07]; combined with the thin-shell estimate of Guédon and Milman [GM11], this improved the record to $\psi_n \lesssim n^{5/12}$. A decisive change came with Eldan’s stochastic-localization scheme, which connects the spectral gap to the evolution of the covariance matrix and relates KLS to thin-shell estimates up to logarithmic losses [Eld13]. With the best thin-shell information then available, Eldan obtained $\psi_n \lesssim n^{1/3} \sqrt{\log n}$. Lee and Vempala subsequently obtained the power bound $\psi_n \lesssim n^{1/4}$ by a streamlined stochastic-localization process and a sharper control of the covariance potential [LV24].

The next advances removed every fixed power of the dimension. Chen obtained the first subpolynomial bound, $\psi_n \leq \exp(C \sqrt{\log n \log \log n})$ [Che21]. Klartag and Lehec then established the first polylogarithmic bound, $\psi_n \lesssim \log^5 n$ [KL22]. Jambulapati, Lee, and Vempala refined the relevant potential argument and improved the exponent to 3.2226 [JLV22]. Klartag’s improved log-concave Lichnerowicz inequality produced the much sharper estimate $\psi_n \lesssim \sqrt{\log n}$ [Kla23].

For $n \geq 2$, the best previous general estimate is due to Letwin [Let26]:

$$\psi_n \lesssim \log^{1/4} n, \quad \text{and hence} \quad \sup_{\substack{\mu \text{ isotropic and log-concave} \\ \text{on } \mathbb{R}^n}} C_P(\mu) \lesssim \sqrt{\log n}. \quad (3)$$

A principal new input is a dimension-free Poincaré inequality for quadratic forms. If X is isotropic and log-concave and M is symmetric, then

$$\text{Var}[\langle MX, X \rangle] \leq 2 \mathbb{E}[\|\nabla \langle MX, X \rangle\|^2] = 8 \text{tr}[M^2]. \quad (4)$$

The first step is Letwin’s quadratic-form inequality, and the equality uses $\nabla \langle MX, X \rangle = 2MX$ and isotropy. This is sometimes described as KLS for quadratic forms; it is not the full KLS conjecture, which concerns all locally Lipschitz functions. To see how Eq. (4) yields the stated general bound, introduce Eldan’s third-moment parameter $\kappa_n := \sup_{\mu} \sup_{\theta \in S^{n-1}} \|\mathbb{E}_{\mu}[\langle X, \theta \rangle X \otimes X]\|_{\text{HS}}$, where the first supremum is over isotropic log-concave laws. Applying Eq. (4) to the matrix inside this definition gives $\kappa_n \leq 2\sqrt{2}$. Combining this with Klartag’s improved Lichnerowicz and stochastic-transfer estimates gives $C_P(\mu) \lesssim \kappa_n \sqrt{\log n}$; the comparison Eq. (2) then yields the exponent 1/4. Moment measures and Stein kernels provide the mechanism behind Letwin’s quadratic inequality; see [Fat19, CFP19] for foundational results in that direction.

For convenience, Table 1 summarizes these selected milestones, all converted to the reciprocal-Cheeger normalization $\psi_{\mu} := h_{\mu}^{-1}$ used throughout this paper.

Dimension-free cases. Several substantial families satisfy KLS with a universal constant. The Gaussian law is the basic model. Poincaré constants tensorize. Every isotropic log-concave law on the line has a universal Poincaré constant; hence products of such laws satisfy KLS [BH97]. The Brascamp–Lieb inequality gives $C_P(e^{-V} dx) \leq \alpha^{-1}$ whenever $\nabla^2 V \succeq \alpha I$ with $\alpha > 0$ [BL76]. This yields a universal dimension-free bound for isotropic laws whose curvature lower bound α is bounded below by a positive universal constant. Dimension-free estimates are also known for ℓ_p^n -balls with $1 \leq p \leq \infty$ [Sod08, LW08, KM14], for the regular simplex [BW09], and for radial log-concave measures and several bodies of revolution [Bob03, Hue11]. Kolesnikov and Milman proved KLS for certain levels of generalized Orlicz balls under a mild growth hypothesis; without that hypothesis their result has a logarithmic loss [KM18]. These special cases are structurally important, but none currently supplies a reduction for arbitrary log-concave measures. In particular, a dimension-free KLS estimate remains open for the full class of unconditional convex bodies. The older logarithmic Poincaré bounds for that class are subsumed by the general bounds above [Kla09, KL25b].

Year	Author(s)	References	Upper bound for ψ_n
1995	Kannan–Lovász–Simonovits	[KLS95]	$\lesssim \sqrt{n}$
2011	Bobkov; Guédon–Milman	[Bob07, GM11]	$\lesssim n^{5/12}$
2013	Eldan	[Eld13]	$\lesssim n^{1/3} \sqrt{\log n}$
2017/2024	Lee–Vempala	[LV24]	$\lesssim n^{1/4}$
2021	Chen	[Che21]	$\leq \exp(C\sqrt{\log n \log \log n})$
2022	Klartag–Lehec	[KL22]	$\lesssim \log^5 n$
2022	Jambulapati–Lee–Vempala	[JLV22]	$\lesssim \log^{3.2226} n$
2023	Klartag	[Kla23]	$\lesssim \sqrt{\log n}$
2026	Letwin	[Let26]	$\lesssim \log^{1/4} n$
2026	This work	Theorem 1.1	$\lesssim \log^{1/6} n$

Table 1: General upper bounds for the KLS constant ψ_n .

Slicing and thin shell. KLS should be distinguished from two neighboring conjectures with which it has historically shared methods. Bourgain’s slicing conjecture asks for a universal bound on the isotropic constant of a volume-one convex body, equivalently for a dimension-free lower bound on the volume of a suitable central hyperplane section [Bou86]. The known implication chain is dimension-free KLS $\implies$ thin shell $\implies$ slicing [EK11]. No converse implication is known without losses. Guan first obtained the near-constant estimate $L_n \lesssim \log \log n$ [Gua24]; Klartag and Lehec then completed an affirmative resolution using Guan’s covariance bound [KL25a]. Bizeul gave a different proof through small-ball estimates [Biz25]. These results solve slicing, not KLS.

The thin-shell conjecture asks whether every isotropic log-concave random vector satisfies

$$\mathbb{E}[(|X| - \sqrt{n})^2] \leq C. \quad (5)$$

To see that KLS implies Eq. (5), put $R := |X|$. Isotropy gives $\mathbb{E}[R^2] = n$, and

$$\mathbb{E}[(R - \sqrt{n})^2] = \frac{2\sqrt{n}}{\sqrt{n} + \mathbb{E}[R]} \text{Var}[R] \leq 2 \text{Var}[R] \leq 2C_P(\mu).$$

The equality follows by expanding both sides and using $\mathbb{E}[R^2] = n$. The first inequality uses $\mathbb{E}[R] \geq 0$, and the second applies the Poincaré inequality to the 1-Lipschitz function $x \mapsto |x|$. Eldan proved a converse only up to polylogarithmic losses [Eld13]; a dimension-free converse remains unknown. Klartag and Lehec proved Eq. (5) by parallel coupling, thereby confirming the thin-shell conjecture [KL25c]. Consequently, as of this writing, slicing and thin shell are theorems, whereas the dimension-free KLS conjecture remains open. Their proofs overlap with KLS work through stochastic localization, covariance processes, and spectral potentials, but their logical conclusions are different. Guan also established exponential-type tail estimates for the localized covariance process [Gua25].

1.1 Our Result

We state our result as follows.

Theorem 1.1 (Informal version of Theorem 3.1). *There are universal constants $C, C' > 0$ such that every isotropic log-concave probability measure μ on $\mathbb{R}^n$, with $n \geq 2$, satisfies*

$$C_P(\mu) \leq C \log^{1/3} n, \quad \psi_n \leq C' \log^{1/6} n.$$

In particular, every locally Lipschitz function f with finite second moment and finite Dirichlet energy satisfies

$$\mathrm{Var}_\mu[f] \leq C \log^{1/3} n \mathbb{E}_\mu[|\nabla f|^2].$$

The best previous general bound in Eq. (3) is due to Letwin [Let26]. Our bound in Theorem 1.1 improves this by a factor of $\log^{1/12} n$, reducing the logarithmic exponent from $1/4$ to $1/6$.

2 Preliminaries

Section 2.1 introduces the notation for vectors and matrices, probabilistic quantities, derivatives, and asymptotic bounds. Section 2.2 states the quadratic-form inequality, the Cheeger–Poincaré comparison, and the bounded-witness lemma.

2.1 Notations

We work with real vectors and matrices. For $x, y \in \mathbb{R}^n$, write $\langle x, y \rangle := x^\top y$ and $|x| := \sqrt{\langle x, x \rangle}$, and set $S^{n-1} := \{x \in \mathbb{R}^n : |x| = 1\}$. We denote the identity matrix by I_n , or by I when the dimension is clear, and the transpose of a matrix M by $M^\top$.

For $M \in \mathbb{R}^{n \times n}$, define $\mathrm{tr}[M] := \sum_{i=1}^n M_{ii}$, $\|M\|_{\mathrm{op}} := \sup_{|x|=1} |Mx|$, and $\|M\|_{\mathrm{HS}} := \sqrt{\mathrm{tr}[M^\top M]}$. The Hilbert–Schmidt inner product is $\langle M, N \rangle_{\mathrm{HS}} := \mathrm{tr}[M^\top N]$; for a finite tensor, its Hilbert–Schmidt norm is the square root of the sum of the squares of its entries. For symmetric matrices A, B , we write $A \preceq B$ if $x^\top (B - A)x \geq 0$ for every $x \in \mathbb{R}^n$.

For a probability measure μ , write $\mathbb{E}_\mu[f] := \int f d\mu$ and $\mathrm{Var}_\mu[f] := \mathbb{E}_\mu[(f - \mathbb{E}_\mu[f])^2]$ whenever these quantities are defined. For square-integrable random vectors U, V on this probability space, define $\mathrm{Cov}_\mu(U, V) := \mathbb{E}_\mu[(U - \mathbb{E}_\mu[U])(V - \mathbb{E}_\mu[V])^\top]$ and $\mathrm{Cov}_\mu(U) := \mathrm{Cov}_\mu(U, U)$; scalar arguments are viewed as one-dimensional vectors. We omit the subscript μ when the underlying law is clear, and write $\|f\|_2 := \sqrt{\mathbb{E}_\mu[f^2]}$ for the real $L^2(\mu)$ norm. A function is centered if $\mathbb{E}_\mu[f] = 0$, and the law of $X \in \mathbb{R}^n$ is isotropic if $\mathbb{E}[X] = 0$ and $\mathrm{Cov}(X) = I_n$.

For a scalar function f on $\mathbb{R}^n$, $\partial_i f$ denotes its derivative in the i th coordinate, ∇f its gradient, and $D^2 f$ its Hessian. We write $\Delta f := \sum_{i=1}^n \partial_i^2 f$ and $\mathrm{Lip}(f) := \sup_{x \neq y} |f(x) - f(y)|/|x - y|$. Derivatives are interpreted weakly when the function is not smooth and the relevant Sobolev regularity is specified.

Throughout, $\log$ denotes the natural logarithm. The symbols $C, c > 0$ denote universal constants, whose values may change from one occurrence to the next, unless a dependence is stated explicitly. For nonnegative quantities a, b , the notation $a \lesssim b$ means $a \leq Cb$ for a universal constant C , and $a \asymp b$ means both $a \lesssim b$ and $b \lesssim a$.

2.2 Tools from previous work

We state several tools from previous work that will be used in Section 3. The numbered citations below refer to Letwin’s arXiv version 2607.24164v1 and Klartag–Lehec’s arXiv version 2406.01324v2.

The first lemma controls quadratic fluctuations under isotropic log-concave laws. It supplies the input for the affine-moment and covariance-noise estimates in Section 3.

Lemma 2.1 (Quadratic forms, [Let26, Theorem 1.2]). *Let X have an isotropic log-concave law μ on $\mathbb{R}^n$. For every real symmetric matrix $B \in \mathbb{R}^{n \times n}$,*

$$\mathrm{Var}_\mu[X^\top B X] \leq 8 \mathrm{tr}[B^2].$$

The next comparison converts bounds on the Poincaré constant into bounds on the inverse Cheeger constant. We use the explicit constants recorded in the remark following Corollary 21 of Klartag and Lehec [KL25b].

Lemma 2.2 (Cheeger–Poincaré comparison, [KL25b, Corollary 21 and the following remark]). *For a log-concave probability measure μ on $\mathbb{R}^n$, let $C_P(\mu)$ be its Poincaré constant and let ψ_μ be its inverse Cheeger constant. Then*

$$C_P(\mu) \leq 4\psi_\mu^2, \quad \psi_\mu^2 \leq \pi C_P(\mu).$$

We state the bounded-witness result in its centered, unit-variance normalization. Its uniform bound will control the contribution of localization paths that exit the covariance window. If a witness is initially bounded only almost everywhere, clipping it to its essential range preserves its values almost everywhere and its Lipschitz constant while making the bound global; centering and variance normalization then give the form below.

Lemma 2.3 (Normalized bounded witness, [KL25b, Theorem 20 and Corollary 21]). *There is a universal constant $B_0 \geq 1$ with the following property. For every log-concave probability measure μ on $\mathbb{R}^n$ with $0 < C_P(\mu) < \infty$, set $k := C_P(\mu)$. Then there exists a bounded Lipschitz function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfying*

$$\mathbb{E}_\mu[f] = 0, \quad \mathbb{E}_\mu[f^2] = 1, \quad \sup_{x \in \mathbb{R}^n} |f(x)| \leq B_0, \quad \text{Lip}(f) \leq B_0/\sqrt{k}.$$

3 Three-generation estimates for the KLS bound

Section 3.1 fixes the Poincaré and Cheeger normalizations, states the main theorem, recalls the external inequalities, and introduces the generator and the regular class of laws. Section 3.2 constructs families of centered normalized derivatives and tracks their total squared norm, Dirichlet energy, and loss from centering. It also derives the symmetric affine moment bound from the quadratic-form inequality. Section 3.3 compares two generations of normalized derivatives with an ordinary Hessian tensor. Spectral excess controls the discrepancy, yielding the three-generation loss estimate without assuming that normalized derivatives commute. Section 3.4 applies that loss estimate in a finite induction to obtain the cubic curvature bound $aC_P(\mu)^3 \leq 288$. Section 3.5 constructs Gaussian stochastic localization and proves simultaneous upper and lower covariance control by a stopped matrix martingale estimate. Section 3.6 selects a localized law that retains a bounded Lipschitz witness, whitens it, and applies the cubic estimate to obtain $C_P(\mu) \lesssim (\log(en))^{1/3}$ for regular starting laws. Section 3.7 justifies the operator domains, Bochner identity, and first eigenfunction, then passes to arbitrary isotropic log-concave laws by approximation and derives the Cheeger exponent $1/6$.

3.1 Normalization and the main conclusion

The Poincaré constant is the best k in

$$\text{Var}_\mu[f] \leq k \mathbb{E}_\mu[|\nabla f|^2].$$

An isotropic law has mean zero and covariance I . Let h_μ be the Cheeger constant defined using perimeter divided by $\min\{\mu(A), 1 - \mu(A)\}$, and set $\psi_\mu := h_\mu^{-1}$ and $\psi_n := \sup_\mu \psi_\mu$, over isotropic log-concave laws on $\mathbb{R}^n$. Lemmas 2.1 and 2.2 give the two inequalities used below:

$$\text{Var}_\mu[X^\top B X] \leq 8 \text{tr}[B^2] \quad (B = B^\top, \mu \text{ isotropic log-concave}), \quad (6)$$

$$\psi_\mu^2 \leq \pi C_P(\mu). \quad (7)$$

In addition, Lemma 2.3 supplies a universal $B_0 \geq 1$ such that every log-concave law with $0 < C_P(\mu) < \infty$ admits, with $k := C_P(\mu)$, a bounded Lipschitz function f satisfying

$$\mathbb{E}_\mu[f] = 0, \quad \mathbb{E}_\mu[f^2] = 1, \quad |f| \leq B_0, \quad \text{Lip}(f) \leq B_0/\sqrt{k}. \quad (8)$$

To justify this normalization, take the bounded 1-Lipschitz witness g from [KL25b, Theorem 20] and set $\sigma^2 := \text{Var}_\mu[g]$. There are universal constants $c_1, C_1 > 0$ such that $\sigma^2 \geq c_1 \psi_\mu^2$ and $\|g\|_\infty^2 \leq C_1 \sigma^2$. If the supremum bound is initially essential, clip g to its essential range; this preserves its values almost everywhere and its Lipschitz constant and makes the bound global. Cheeger's direction of Lemma 2.2 gives $\psi_\mu^2 \geq k/4$, hence $\sigma^2 \geq c_1 k/4 > 0$. Define $f := (g - \mathbb{E}_\mu[g])/\sigma$. Then $\mathbb{E}_\mu[f] = 0$ and $\mathbb{E}_\mu[f^2] = 1$, while

$$\|f\|_\infty \leq 2\|g\|_\infty/\sigma \leq 2\sqrt{C_1}, \quad \text{Lip}(f) \leq 1/\sigma \leq 2/\sqrt{c_1 k}.$$

In the first chain, centering gives the first inequality and the supremum bound for g gives the second. In the second chain, the 1-Lipschitz property gives the first inequality and the variance lower bound gives the second. Thus $B_0 := \max\{1, 2\sqrt{C_1}, 2/\sqrt{c_1}\}$ gives Eq. (8).

The next theorem states the conclusion of the three-generation argument. The proof first establishes a cubic relation between the Poincaré constant and a lower curvature bound, and then transfers that estimate to general laws by stochastic localization and approximation. The Cheeger comparison converts the one-third logarithmic exponent for the Poincaré constant into the one-sixth exponent for the KLS constant.

Theorem 3.1 (Main result, formal version of Theorem 1.1). *Using the external inputs above, there are universal constants C, C' such that, for every isotropic log-concave probability measure μ on $\mathbb{R}^n$,*

$$C_P(\mu) \leq C(\log(en))^{1/3}, \quad \psi_n \leq C'(\log(en))^{1/6}.$$

The proof first treats regular laws $d\mu = e^{-W}dx$, where W is smooth and for some law-dependent $0 < a \leq b < \infty$,

$$aI \preceq D^2W \preceq bI. \quad (9)$$

Neither a nor b is assumed uniform. Let

$$H := -\Delta + \nabla W \cdot \nabla, \quad \lambda := C_P(\mu)^{-1}, \quad P_+h := h - \mathbb{E}_\mu[h].$$

H is the nonnegative self-adjoint operator associated to the Dirichlet form $\mathbb{E}[\nabla f \cdot \nabla g]$ on $L^2(\mu)$; inverse powers below act only on the centered subspace. Lemma 3.10 gives a positive gap λ , a real unit first eigenfunction, and the inverse-square-root domain properties used below. Lemma 3.9 gives the integrated Bochner identity on $\text{Dom}(H)$:

$$\|Hg\|_2^2 = \mathbb{E}[\|D^2g\|_{\text{HS}}^2] + \mathbb{E}[(\nabla g)^\top D^2W \nabla g]. \quad (10)$$

3.2 Finite families and their exact losses

For a finite family of real centered form-domain functions $\mathcal{F} := (h_\alpha)$ define its successor by the finite family

$$g_\alpha := H^{-1/2}h_\alpha, \quad u_{i,\alpha} := P_+\partial_i g_\alpha.$$

Set

$$v := \sum_{\alpha} \|h_{\alpha}\|_2^2, \quad e := \sum_{\alpha} \|H^{1/2}h_{\alpha}\|_2^2, \quad \varepsilon := e - \lambda v \geq 0.$$

Primes will refer to the successor family, and $p := v - v'$. Every successor is again in the form domain: Lemma 3.10 gives $g_{\alpha} \in \text{Dom}(H)$, and Lemma 3.9 gives $D^2g_{\alpha} \in L^2(\mu)$ and $\partial_i g_{\alpha}$ in the form domain.

The successor operation can reduce the total variance when the derivatives are centered. The next lemma quantifies this loss and the simultaneous decrease in Dirichlet energy due to curvature. These estimates provide the recurrence for the spectral excess used in the finite iteration.

Lemma 3.2. *For every such family,*

$$0 \leq p \leq \min\{v, e\}, \quad e' \leq e - av, \quad \varepsilon' \leq \varepsilon + \lambda p - av. \quad (11)$$

Proof. Isotropy says that the coordinate functions are orthonormal in L^2 . Thus the coordinate map $Lz := \mathbb{E}[Xz]$ satisfies $|Lz| \leq \|z\|_2$. The form-domain testing identity in Lemma 3.9, applied with $\phi := X_i$, gives

$$\mathbb{E}[\nabla g] = LHg = LH^{1/2}h, \quad \mathbb{E}[|\nabla g|^2] = \|h\|_2^2.$$

Centering each derivative therefore loses exactly

$$p = \sum_{\alpha} |\mathbb{E}[\nabla g_{\alpha}]|^2 = \sum_{\alpha} |LH^{1/2}h_{\alpha}|^2 \leq e.$$

The sum-of-squares formula gives $p \geq 0$. Also, $p = v - v' \leq v$ because v' is a sum of squared L^2 norms and is therefore nonnegative. Summing Eq. (10) yields

$$e' = \sum_{\alpha} \mathbb{E}[\|D^2g_{\alpha}\|_{\text{HS}}^2] \leq e - a \sum_{\alpha} \mathbb{E}[|\nabla g_{\alpha}|^2] = e - av.$$

Subtracting $\lambda v' = \lambda(v - p)$ proves the last inequality. $\square$

We next use the quadratic-form inequality to control the symmetric part of the correlation matrix between the coordinates and a gradient. The resulting bound is expressed in terms of the norm of the generator applied to the function. Applied to each member of a successor family, the lemma supplies the symmetric tensor estimate needed below. For a real square matrix M , we use the averaged symmetrization $\text{Sym } M := (M + M^{\top})/2$.

Lemma 3.3. *If $g \in \text{Dom}(H)$, then*

$$\|\text{Sym } \mathbb{E}[X(\nabla g)^{\top}]\|_{\text{HS}}^2 \leq 2\|Hg\|_2^2.$$

Proof. For every symmetric B , quadratic testing gives

$$2\langle B, \text{Sym } \mathbb{E}[X(\nabla g)^{\top}] \rangle_{\text{HS}} = \mathbb{E}[(X^{\top}BX - \text{tr}[B])Hg].$$

The right side has absolute value at most $\sqrt{8}\|B\|_{\text{HS}}\|Hg\|_2$ by Lemma 2.1, in the form of Eq. (6). Take the supremum over symmetric B of Hilbert–Schmidt norm one. The polynomial form-domain assertion and testing identity in Lemma 3.9 justify these tests. $\square$

3.3 The joint tensor estimate

The next lemma is an algebraic step that controls a full tensor using a symmetrized part and an error term. Symmetry in the last two indices is essential for the exact identity. In the application, that symmetry comes from commuting weak mixed derivatives, while the error accounts for the intervening inverse square root of the generator. For a real three-index tensor S , our symmetrizations are the averages

$$(\text{Sym}_{12} S)_{ijk} := \frac{S_{ijk} + S_{jik}}{2}, \quad (\text{Sym}_{123} S)_{ijk} := \frac{S_{ijk} + S_{ikj} + S_{jik} + S_{jki} + S_{kij} + S_{kji}}{6}.$$

Lemma 3.4. *If a real three-index tensor S obeys $S_{ijk} = S_{ikj}$, then*

$$4\|\text{Sym}_{12} S\|_{\text{HS}}^2 = \|S\|_{\text{HS}}^2 + 3\|\text{Sym}_{123} S\|_{\text{HS}}^2.$$

Consequently, for any tensor T and such an S ,

$$\|T\|_{\text{HS}} \leq 2\|\text{Sym}_{12} T\|_{\text{HS}} + 3\|T - S\|_{\text{HS}}. \quad (12)$$

The same assertion holds with the norms summed in quadrature over any finite ancestor index.

Proof. Write $N := \sum S_{ijk}^2$ and $c := \sum S_{ijk}S_{jik}$. Symmetry in the last two indices and relabeling give $\|\text{Sym}_{12} S\|^2 = (N + c)/2$ and $\|\text{Sym}_{123} S\|^2 = (N + 2c)/3$. The identity follows. Apply the triangle inequality to $T = S + (T - S)$, then use $\|S\| \leq 2\|\text{Sym}_{12} S\|$ and that symmetrization is a contraction. This proof also works in the direct sum over ancestors. $\square$

Iterate the successor operation, writing $v_r, e_r, \varepsilon_r, p_r$ for the quantities at generation r . In particular p_r is the loss from generation r to generation $r + 1$.

We now combine the quadratic-form estimate with Lemma 3.4 to control variance loss across three successive generations. The two spectral excess terms measure deviations from the bottom spectral scale, so no commutation of differentiation with the inverse square root of the generator is assumed. This is the estimate used to close the finite iteration.

Proposition 3.5. *For every $r \geq 0$,*

$$\sqrt{p_{r+2}} \leq 2\sqrt{2\lambda e_{r+1}} + 3\sqrt{\varepsilon_{r+1}} + \sqrt{\varepsilon_{r+2}}. \quad (13)$$

Proof. Reset the ancestor generation to zero and write

$$g_\alpha := H^{-1/2}h_\alpha, \quad u_{k,\alpha} := P_+\partial_k g_\alpha, \quad g_{k,\alpha} := H^{-1/2}u_{k,\alpha}, \quad z_{j,k,\alpha} := P_+\partial_j g_{k,\alpha}.$$

The u and z families are generations one and two. Define

$$T_{ijk}^\alpha := \mathbb{E}[X_i \partial_j g_{k,\alpha}], \\ S_{ijk}^\alpha := \lambda^{-1/2} \mathbb{E}[X_i \partial_j u_{k,\alpha}] = \lambda^{-1/2} \mathbb{E}[X_i \partial_j \partial_k g_\alpha].$$

Lemma 3.9 gives commuting weak mixed derivatives of g_α and integrable pairings with X_i , so S is symmetric in j, k . This assertion is about S , not T ; no commutation of $H^{-1/2}$ with differentiation is assumed.

Lemma 3.3, applied separately for each k, α , gives

$$\sum_\alpha \|\text{Sym}_{12} T^\alpha\|_{\text{HS}}^2 \leq 2 \sum_{k,\alpha} \|H g_{k,\alpha}\|_2^2 = 2e_1. \quad (14)$$

Coordinate Bessel applied before summing over derivatives gives

$$\begin{aligned}
\sum_{\alpha} \|T^{\alpha} - S^{\alpha}\|_{\text{HS}}^2 &\leq \sum_{k,\alpha} \|\nabla(H^{-1/2} - \lambda^{-1/2})u_{k,\alpha}\|_2^2 \\
&= \lambda^{-1} \sum_{k,\alpha} \|(H^{1/2} - \sqrt{\lambda})u_{k,\alpha}\|_2^2 \\
&\leq \lambda^{-1} \sum_{k,\alpha} \langle u_{k,\alpha}, (H - \lambda)u_{k,\alpha} \rangle = \varepsilon_1/\lambda.
\end{aligned} \tag{15}$$

The last pairing is a quadratic form. The last inequality is the spectral inequality $(\sqrt{x} - \sqrt{\lambda})^2 \leq x - \lambda$ for $x \geq \lambda$. All functions in the first line are in the form domain. In particular, the equality uses the Dirichlet form identity, not derivative commutation.

Applying Lemma 3.4, in the form of Eq. (12), and combining Eq. (14) with Eq. (15) in the ancestor direct sum,

$$(\sum_{\alpha} \|T^{\alpha}\|_{\text{HS}}^2)^{1/2} \leq 2\sqrt{2e_1} + 3\sqrt{\varepsilon_1/\lambda}.$$

By Lemma 3.2, $p_2 = \sum_{j,k,\alpha} |LH^{1/2}z_{j,k,\alpha}|^2$. Since $\mathbb{E}[X] = 0$, $Lz_{j,k,\alpha}$ has entries T_{ijk}^{α} . Split $H^{1/2}z = \sqrt{\lambda}z + (H^{1/2} - \sqrt{\lambda})z$, use the direct-sum triangle inequality, coordinate Bessel, and the same spectral inequality. The result is

$$\sqrt{p_2} \leq \sqrt{\lambda}(\sum_{\alpha} \|T^{\alpha}\|_{\text{HS}}^2)^{1/2} + \sqrt{\varepsilon_2}.$$

This proves Eq. (13). Resetting to any generation proves the stated quantifier. No factor counting coordinates or ancestors occurs. $\square$

The spectral error bound can also be expressed as a commutator estimate. In the notation of the proof, put $R_{k,\alpha} := (H^{-1/2} - \lambda^{-1/2})u_{k,\alpha}$. Ordinary Hessian symmetry gives

$$z_{j,k,\alpha} - z_{k,j,\alpha} = P_+(\partial_j R_{k,\alpha} - \partial_k R_{j,\alpha}),$$

so the Dirichlet calculation in Eq. (15) implies

$$\sum_{\alpha,j,k} \|z_{j,k,\alpha} - z_{k,j,\alpha}\|_2^2 \leq 4 \sum_{\alpha,k} \|\nabla R_{k,\alpha}\|_2^2 \leq 4\varepsilon_1/\lambda.$$

This quantity is allowed to be nonzero. No exact symmetry of the normalized derivatives is needed by the proof.

3.4 Finite iteration and cubic curvature dependence

The next theorem is the curvature estimate at the center of the three-generation argument. Starting from a first eigenfunction, the proof iterates the variance and energy estimates while retaining a fixed fraction of the total variance. If the cubic relation failed, the spectral excess would become negative after finitely many generations.

Theorem 3.6 (Cubic curvature estimate). *Every regular isotropic law satisfying Eq. (6) and $D^2W \succeq aI$ obeys*

$$aC_P(\mu)^3 \leq 288.$$

Proof. By Lemma 3.10, choose a real unit first eigenfunction f of H and start the successor iteration from it. Thus $v_0 = 1$, $e_0 = \lambda$, and $\varepsilon_0 = 0$. Eq. (11) gives, for every finite generation,

$$\begin{aligned} 0 \leq p_r \leq e_r \leq \lambda, \quad v_r &\geq 1 - r\lambda, \\ 0 \leq \varepsilon_{r+1} \leq \varepsilon_r + \lambda p_r - av_r. \end{aligned}$$

In particular $\varepsilon_1 \leq \lambda^2$ and $\varepsilon_2 \leq 2\lambda^2$. The first step also gives $a \leq \lambda^2$:

$$\lambda(1 - p_0) \leq e_1 \leq \lambda - a, \quad p_0 \leq \lambda.$$

If $\lambda > 1/16$, then $a \leq \lambda^2 < 16\lambda^3$, as required. For $\lambda \leq 1/16$, suppose for contradiction that $a \geq 288\lambda^3$ and set

$$m := \lceil \frac{8\lambda^2}{a} \rceil + 2.$$

Then $m\lambda \leq 8\lambda^3/a + 3\lambda \leq 1/36 + 3/16 = 31/144 < 1/4$, so $v_r \geq 3/4$ for $0 \leq r \leq m$. Inductively suppose the excesses through generation $r \geq 2$ are at most $2\lambda^2$. Proposition 3.5, with initial generation $r - 2$, yields

$$\sqrt{p_r} \leq (2\sqrt{2} + 3\sqrt{2} + \sqrt{2})\lambda = 6\sqrt{2}\lambda, \quad p_r \leq 72\lambda^2.$$

For $2 \leq r < m$, it follows that

$$\varepsilon_{r+1} \leq \varepsilon_r + 72\lambda^3 - 3a/4 \leq \varepsilon_r - a/2.$$

This preserves the inductive upper bound, starting from generations one and two. At generation m it instead forces

$$0 \leq \varepsilon_m \leq 2\lambda^2 - (m - 2)a/2 \leq -2\lambda^2,$$

a contradiction. The argument uses only finitely many function families. $\square$

Inputs to the argument. The cubic curvature estimate uses centering and isotropy for Bessel's inequality applied to the coordinate functions, the quadratic-form inequality in Eq. (6) for the symmetric affine-moment bound, the integrated Bochner identity with the curvature lower bound, and the spectral gap. Lemmas 3.9 and 3.10 justify the operator domains and the first eigenfunction used in the finite iteration. The transfer to general laws additionally uses the bounded witness from Lemma 2.3, the Cheeger–Poincaré comparison from Lemma 2.2, and the localization and approximation arguments below; it does not invoke a previous logarithmic KLS bound.

3.5 Localization with simultaneous covariance bounds

We first give the Gaussian observation construction and justify its identities for all tests used below. The argument applies to bounded Borel functions and to functions of polynomial growth, so it includes the coordinate products needed for the covariance equation. This construction is standard; see [KL25b, Section 6.2 and Lemma 54].

Lemma 3.7 (Gaussian posterior identities). *Let μ be a regular isotropic law satisfying Eq. (9). Let $X \sim \mu$ be independent of a standard Brownian motion B_t , put $Y_t := tX + B_t$, and let $\mathcal{F}_t$ be the usual augmentation of the observation filtration generated by $(Y_s)_{0 \leq s \leq t}$. The conditional law μ_t of X given $\mathcal{F}_t$ is*

$$d\mu_t(x) = Z_t^{-1} \exp(Y_t \cdot x - t|x|^2/2) d\mu(x), \quad Z_t := \int \exp(Y_t \cdot x - t|x|^2/2) d\mu(x).$$

Write $m_t := \mathbb{E}_{\mu_t}[X]$ and $A_t := \text{Cov}_{\mu_t}(X)$. Then $W_t := Y_t - \int_0^t m_s ds$ is a standard Brownian motion in $(\mathcal{F}_t)$. For every Borel q of polynomial growth, $\mathbb{E}_{\mu_t}[q]$ is a square-integrable martingale and

$$d\mathbb{E}_{\mu_t}[q] = \text{Cov}_{\mu_t}(q, X) \cdot dW_t. \quad (16)$$

In particular, $dm_t = A_t dW_t$ and

$$dA_t = \sum_j U_{j,t} dW_{j,t} - A_t^2 dt, \quad U_{j,t} := \mathbb{E}_{\mu_t}[(X_j - m_{j,t})(X - m_t)(X - m_t)^\top]. \quad (17)$$

Every posterior has a positive smooth log-concave density with potential Hessian $D^2W + tI$, bounded between $(a + t)I$ and $(b + t)I$.

Proof. For fixed x , the law of $(sx + B_s)_{0 \leq s \leq t}$ has density $\exp(x \cdot Y_t - t|x|^2/2)$ relative to Wiener measure on paths up to time t . Indeed, multiplying the Gaussian likelihood ratios of the independent increments along any finite partition gives this endpoint expression. Cylinder sets generate the path sigma-field, so the same identity holds on that sigma-field. Integrating against $\mu(dx)$ and applying Bayes' formula proves the posterior formula and $\mathbb{E}_{\mu_t}[q] = \mathbb{E}[q(X) \mid \mathcal{F}_t]$. Strong convexity gives Gaussian tails for μ , hence all polynomial moments are finite; conditional Jensen therefore makes these martingales square integrable.

For $0 \leq s < t$, Brownian increments after s are independent of X and of the Brownian path up to s . Moreover, the tower property gives $\mathbb{E}[m_r \mid \mathcal{F}_s] = \mathbb{E}[X \mid \mathcal{F}_s] = m_s$ for $r \geq s$. Consequently $\mathbb{E}[W_t - W_s \mid \mathcal{F}_s] = 0$. All terms are integrable because $\mathbb{E}[|m_r|^2] \leq \mathbb{E}[|X|^2]$ by conditional Jensen. The process W_t is continuous and its quadratic covariation is tI , since subtracting the finite-variation drift does not change quadratic covariation. Lévy's characterization now shows that W_t is Brownian in the observation filtration.

For the stochastic differential, define

$$I_q(t, y) := \int q(x) \exp(y \cdot x - t|x|^2/2) d\mu(x).$$

Gaussian tails permit differentiation under this integral locally uniformly in (t, y) , including a neighborhood of $t = 0$. In particular $\partial_t I_q + \Delta_y I_q/2 = 0$ and $\nabla_y I_q = I_{qx}$, where the vector integral I_{qx} is defined componentwise. Since $Z_t = I_1(t, Y_t)$, Itô's formula gives $dI_q(t, Y_t) = \nabla_y I_q(t, Y_t) \cdot dY_t$ and $dZ_t = Z_t m_t \cdot dY_t$. The quotient rule, including its quadratic-covariation terms, yields

$$d(I_q(t, Y_t)/Z_t) = \text{Cov}_{\mu_t}(q, X) \cdot (dY_t - m_t dt) = \text{Cov}_{\mu_t}(q, X) \cdot dW_t.$$

The first equality uses $I_{qx}/Z_t - (I_q/Z_t)m_t = \text{Cov}_{\mu_t}(X, q)$; the second uses the definition of W_t . These calculations may first be stopped when $|Y_t|$ reaches R . For every finite T , conditional Cauchy–Schwarz followed by Cauchy–Schwarz and conditional Jensen gives

$$\mathbb{E}_{\text{loc}} \left[\int_0^T |\text{Cov}_{\mu_t}(q, X)|^2 dt \right] \leq T(\mathbb{E}_{\mu}[q^4] \mathbb{E}_{\mu}[|X|^4])^{1/2} < \infty.$$

Here conditional Cauchy–Schwarz bounds the squared covariance by $\mathbb{E}_{\mu_t}[q^2] \mathbb{E}_{\mu_t}[|X|^2]$, and the other two inequalities bound the expectation of this product by the displayed fourth moments. Continuity of Y_t on finite intervals and the Itô isometry now let $R \rightarrow \infty$, proving Eq. (16) for the unbounded tests as well.

Apply Eq. (16) to x_i and $x_i x_k$. The first choice gives $dm_t = A_t dW_t$. Writing $Q_t := \mathbb{E}_{\mu_t}[XX^\top]$ and $A_t = Q_t - m_t m_t^\top$, the product rule contributes the drift $-dm_t dm_t^\top = -A_t^2 dt$. For each Brownian coordinate j , its remaining coefficient is

$$\mathbb{E}_{\mu_t}[XX^\top(X_j - m_{j,t})] - (A_t e_j) m_t^\top - m_t (A_t e_j)^\top = U_{j,t},$$

where e_j is the j th coordinate vector; expanding $X = (X - m_t) + m_t$ proves the equality. This establishes Eq. (17). Finally, the posterior potential is $W(x) - Y_t \cdot x + t|x|^2/2 + \log Z_t$, which gives the asserted Hessian bounds. $\square$

To use the curvature estimate after localization, the covariance must remain bounded both above and below. The next lemma bounds the probability of leaving a fixed neighborhood of the identity throughout a short time interval. For a sufficiently small constant multiple of $1/\log(en)$, this gives the event on which whitening preserves a curvature lower bound of order the elapsed localization time.

Lemma 3.8. *Let τ be the first exit of A_t from $(I/2, 2I)$. For $0 < t \leq 1/16$,*

$$\Pr[\tau \leq t] \leq 2n \exp(-1/(2048t)). \quad (18)$$

Proof. For any centered log-concave Y with positive covariance A , applying Lemma 2.1 after whitening gives, for every symmetric D ,

$$\text{Var}[Y^\top D Y] \leq 8\|A^{1/2} D A^{1/2}\|_{\text{HS}}^2 \leq 8\|A\|_{\text{op}}^2 \|D\|_{\text{HS}}^2.$$

Set $S_z := \mathbb{E}[(z \cdot Y) Y Y^\top]$. Trace duality and Cauchy–Schwarz give

$$|\langle D, S_z \rangle_{\text{HS}}| = |\mathbb{E}[(z \cdot Y)(Y^\top D Y - \text{tr}[DA])]| \leq \sqrt{8z^\top A z} \|A\|_{\text{op}} \|D\|_{\text{HS}}.$$

Taking the supremum over symmetric D of unit norm shows $\|S_z\|_{\text{HS}}^2 \leq 8\|A\|_{\text{op}}^2 z^\top A z$. Symmetry of the third moment tensor gives $\sum_j |U_j z|^2 = \|S_z\|_{\text{HS}}^2$. Apply this to each centered posterior to conclude $\sum_j U_{j,s}^2 \leq 64I$ before τ .

Set $M_u := \int_0^{u \wedge \tau} \sum_j U_{j,s} dW_{j,s}$. Fix $\theta > 0$. For symmetric matrices M, T ,

$$D^2 \text{tr}[e^{\theta M}][T, T] \leq \theta^2 \text{tr}[e^{\theta M} T^2].$$

To verify it, diagonalize M . The coefficient of T_{ij}^2 on the left is θ^2 times the logarithmic mean of $e^{\theta m_i}$ and $e^{\theta m_j}$; this is at most their arithmetic mean, which gives the right side after summation. Itô's formula now makes $Z_u := e^{-32\theta^2 u} \text{tr}[e^{\theta M_u}]$ a nonnegative local supermartingale, hence a supermartingale. At the first upper eigenvalue crossing of level r before time t , $Z_u \geq e^{\theta r - 32\theta^2 t}$. Optional stopping, $Z_0 = n$, optimization in $\theta > 0$, and the same argument for $-M$ yield

$$\Pr[\sup_{u \leq t} \|M_u\|_{\text{op}} \geq r] \leq 2n \exp(-r^2/(128t)).$$

On $\tau \leq t$, Eq. (17) gives $M_\tau = A_\tau - I + \int_0^\tau A_s^2 ds$. An upper exit forces an eigenvalue at least one. A lower exit, tested against its unit eigenvector, forces an eigenvalue at most $-1/2 + 4t \leq -1/4$. Use $r := 1/4$ to conclude. $\square$

3.6 A surviving witness and the exponent

Let $k := C_P(\mu)$ and choose f from Lemma 2.3, so that Eq. (8) holds. Let τ be the exit time from Lemma 3.8. Lemma 3.7 shows that the posterior expectations of f and f^2 are bounded martingales. For $V_s := \text{Var}_{\mu_{s \wedge \tau}}[f]$, Eq. (16), Itô's formula, and optional stopping give

$$F(t) := \mathbb{E}_{\text{loc}}[V_t] = 1 - \int_0^t \mathbb{E}_{\text{loc}}[\mathbf{1}_{\{s < \tau\}} |\text{Cov}_{\mu_s}(X, f)|^2] ds.$$

Before exit, directional Cauchy–Schwarz gives $|\text{Cov}_{\mu_s}(X, f)|^2 \leq \|A_s\|_{\text{op}} V_s \leq 2V_s$. Therefore $F'(t) \geq -2F(t)$ almost everywhere, and

$$F(t) \geq e^{-2t}, \quad 0 \leq V_t \leq B_0^2. \quad (19)$$

Choose

$$c_0 := \frac{1}{2048 \log(16B_0^2)}, \quad t := \frac{c_0}{\log(en)}.$$

Since $B_0 \geq 1$ and $\log(en) \geq 1$, we have $0 < t \leq c_0 < 1/16$. Writing $q := 1/(2048c_0) = \log(16B_0^2) > 1$, Lemma 3.8, in the form of Eq. (18), gives $\Pr[\tau \leq t] \leq 2ne^{-1/(2048t)} = 2e^{-q}n^{1-q} \leq 1/(8B_0^2)$. The equality uses $t = c_0/\log(en)$, and the last inequality uses $q > 1$ and $e^{-q} = 1/(16B_0^2)$. Also $e^{-2t} \geq 1 - 2t \geq 7/8$, first by convexity of the exponential and then by $t \leq 1/16$. Using Eq. (19) and subtracting the contribution of the exit event, we obtain

$$\mathbb{E}_{\text{loc}}[\mathbf{1}_{\{\tau > t\}} \text{Var}_{\mu_t}[f]] \geq 3/4.$$

There is a realization with $\tau > t$ and $\text{Var}_{\mu_t}[f] > 1/2$. Fix this realization and its mean m and covariance A , and let η be the law of $A^{-1/2}(X - m)$ under μ_t . The whitened potential has Hessian

$$A^{1/2}(D^2W + tI)A^{1/2} \succeq (t/2)I,$$

and it has a finite upper Hessian bound. The law η is isotropic and log-concave, so Lemma 2.1 supplies Eq. (6) for η . Thus every hypothesis of Theorem 3.6 holds, and it gives $C_P(\eta) \leq (576/t)^{1/3}$. On the other hand $z \mapsto f(A^{1/2}z + m)$ has variance greater than $1/2$ and Dirichlet energy at most $\|A\|_{\text{op}} \text{Lip}(f)^2 \leq 2B_0^2/k$. Hence

$$\frac{k}{4B_0^2} \leq C_P(\eta) \leq (576/t)^{1/3}, \quad k \leq 4B_0^2(576/c_0)^{1/3}(\log(en))^{1/3}.$$

This proves the Poincaré assertion for every regular isotropic starting law. Lemmas 3.11 and 3.12 extend it to all isotropic log-concave laws, as detailed at the end of Section 3.7. There is no hypothesis on the starting law's proximity to an extremizer and no use of the old quarter-power bound in this transfer.

3.7 Domains and approximation

The first two lemmas justify the operator calculations and the initial eigenfunction used in the finite iteration. The last two lemmas provide the approximation and limiting argument needed to extend the regular-law bound to every isotropic log-concave law.

The next lemma makes the integration-by-parts calculations valid on the operator domain, rather than only for compactly supported smooth functions. It also supplies the weak Hessians and polynomial testing identities used in Sections 3.2 and 3.3.

Lemma 3.9 (Operator core and Bochner identity). *Let $d\mu = e^{-W}dx$ be a regular law satisfying Eq. (9), and let H be its nonnegative self-adjoint Dirichlet-form generator on $L^2(\mu)$. Its form domain is the weighted Sobolev space $\{g \in L^2(\mu) : \nabla g \in L^2(\mu)\}$, with weak derivatives, and its kernel consists exactly of the constants. The space $C_c^\infty(\mathbb{R}^n)$ is a graph core for H , and $P_+C_c^\infty(\mathbb{R}^n)$ is a graph core on the centered subspace. For every $g \in \text{Dom}(H)$, all second weak derivatives belong to $L^2(\mu)$, mixed derivatives commute, and Eq. (10) holds. In particular, each $\partial_i g$ belongs to the form domain. Every polynomial belongs to the form domain, and $\langle \phi, Hg \rangle_{L^2(\mu)} = \mathbb{E}_\mu[\nabla \phi \cdot \nabla g]$ for every form-domain test ϕ .*

Proof. On each compact set, the smooth positive weight is bounded above and below by positive constants. Cutoff and mollification therefore identify the closure of C_c^∞ in the form norm with the stated weighted Sobolev space. More explicitly, choose smooth cutoffs χ_R with $0 \leq \chi_R \leq 1$, equal to one on the radius- R ball and zero outside the radius- $2R$ ball, with $|\nabla \chi_R| \leq C/R$ and $|D^2 \chi_R| \leq C/R^2$. For a weighted Sobolev function g , the functions $\chi_R g$ converge to g in the form norm: the terms involving $1 - \chi_R$ vanish by integrability, and $\|g \nabla \chi_R\|_2 \leq (C/R)\|g\|_2$. Each compactly supported member is then approximated by ordinary mollification on a fixed compact neighborhood of its support. The same cutoffs approximate the constant one, which has zero energy. Conversely, a zero-energy function has weak gradient zero and hence is constant on the connected space $\mathbb{R}^n$. This also identifies $\ker H$.

Let $g \in \text{Dom}(H)$. The weak equation $-\Delta g + \nabla W \cdot \nabla g = Hg$ and local elliptic regularity give $g \in H_{\text{loc}}^2$. The product rule gives

$$H(\chi_R g) = \chi_R Hg - 2\nabla \chi_R \cdot \nabla g + g H \chi_R.$$

This identity initially holds distributionally. Its right side is in $L^2(\mu)$, so the defining weak characterization of the generator shows that $\chi_R g \in \text{Dom}(H)$. Since $|\nabla W(x)| \leq |\nabla W(0)| + b|x|$, the function $H\chi_R$ is bounded by a law-dependent constant independent of $R \geq 1$ and is supported in the annulus $R \leq |x| \leq 2R$. Consequently $\chi_R Hg \rightarrow Hg$ in $L^2(\mu)$, $\|\nabla \chi_R \cdot \nabla g\|_2 \rightarrow 0$, and $\|g H \chi_R\|_2 \rightarrow 0$. Thus $\chi_R g \rightarrow g$ in the graph norm $\|g\|_2 + \|Hg\|_2$. Mollifying each compactly supported H^2 function gives smooth compactly supported functions converging in H^2 on a common compact set. The bounded coefficients there imply convergence of their images under H in $L^2(\mu)$. A diagonal choice proves the graph-core assertion. Subtracting the means gives the centered graph core, because $H1 = 0$ and convergence in $L^2(\mu)$ implies convergence of the means.

For a smooth compactly supported g , differentiation gives $\partial_i(Hg) = H(\partial_i g) + \sum_j (\partial_i \partial_j W) \partial_j g$. Integration by parts in each coordinate proves Eq. (10). Let g_j be a graph-core sequence for a general $g \in \text{Dom}(H)$. For a difference $u := g_j - g_k$, the form identity gives

$$\|\nabla u\|_2^2 = \langle u, Hu \rangle_{L^2(\mu)} \leq \|u\|_2 \|Hu\|_2.$$

The equality is the definition of the generator through its form, and the inequality is Cauchy–Schwarz. Hence the gradients converge in $L^2(\mu)$. Applying Eq. (10) to u and using $D^2 W \succeq 0$ shows that the Hessians are Cauchy in $L^2(\mu)$ as well. Their limits are the distributional second derivatives of g . The bound $D^2 W \preceq bI$ passes the curvature term to the limit and proves Eq. (10) on all of $\text{Dom}(H)$. Distributional mixed derivatives commute, and the form-domain characterization now applies to each $\partial_i g$.

Finally, strong convexity gives $W(x) \geq W(0) + \nabla W(0) \cdot x + a|x|^2/2$. Thus every polynomial and its gradient are square integrable, which places all polynomials in the form domain. The asserted testing identity is the defining generator identity on that domain. In the isotropic case, the pairings $\mathbb{E}[X_i \partial_j \partial_k g]$ are also finite by Cauchy–Schwarz. $\square$

The finite iteration begins with a first eigenfunction rather than an approximate minimizer of the Rayleigh quotient. The next lemma supplies this eigenfunction and makes the inverse square root used in the successor operation well defined.

Lemma 3.10 (First eigenfunction and inverse square root). *For every regular law satisfying Eq. (9), the operator H has compact resolvent and a positive first nonzero eigenvalue $\lambda = C_P(\mu)^{-1}$. There is a real $f \in \text{Dom}(H)$ with $\mathbb{E}_\mu[f] = 0$, $\|f\|_2 = 1$, and $Hf = \lambda f$. The operator $H^{-1/2}$ is bounded on the centered subspace. For every centered $h \in \text{Dom}(H^{1/2})$, the function $g := H^{-1/2}h$ belongs to $\text{Dom}(H)$ and satisfies $\|Hg\|_2^2 = \|H^{1/2}h\|_2^2$ and $\mathbb{E}_\mu[|\nabla g|^2] = \|h\|_2^2$.*

Proof. The unitary map $Ug := e^{-W/2}g$ from $L^2(\mu)$ to $L^2(dx)$ transforms H into the Friedrichs realization of $-\Delta + V$, where $V := |\nabla W|^2/4 - \Delta W/2$. This follows by direct differentiation on the core from Lemma 3.9, followed by closure of the forms. Strong convexity gives a unique minimizer x_* of W and $|\nabla W(x)| \geq a|x - x_*|$, while $\Delta W \leq nb$. It follows that $V(x) \geq a^2|x - x_*|^2/4 - nb/2$.

After adding $nb/2 + 1$ to this potential, its form norm controls $\|u\|_2$, $\|\nabla u\|_2$, and $\| |x - x_*| u \|_2$. A sequence bounded in this norm has a subsequence converging in L^2 on each fixed ball by Rellich compactness. Outside the radius- R ball about x_* , its squared L^2 norm is at most $R^{-2} \| |x - x_*| u \|_2^2$, uniformly along the sequence. A diagonal subsequence and this tail estimate give convergence in all of $L^2(dx)$. The form-domain embedding is therefore compact, which gives compact resolvent for H by the unitary transformation.

Lemma 3.9 identifies the kernel as the one-dimensional space of constants. The discrete spectrum consequently has a positive first nonzero eigenvalue, attained on the centered subspace. Denote this eigenvalue temporarily by λ_1 . The Rayleigh variational principle gives $\text{Var}_\mu[q] \leq \lambda_1^{-1} \mathbb{E}_\mu[|\nabla q|^2]$ for every compactly supported smooth q . Apply Lemma 3.12 to the constant sequence $\mu_j := \mu$ with $K := \lambda_1^{-1}$. Its truncation and cutoff argument extends the inequality to every finite-energy locally Lipschitz function and proves $C_P(\mu) \leq \lambda_1^{-1}$; that lemma does not use the present spectral argument. Conversely, approximate a unit first eigenfunction in the form norm by compactly supported smooth functions using Lemma 3.9. Their variances tend to one and their energies tend to λ_1 , so the definition of $C_P(\mu)$ gives $C_P(\mu) \geq \lambda_1^{-1}$. Thus $\lambda_1 = C_P(\mu)^{-1} = \lambda$, with the full test class in the definition of C_P accounted for. Since the operator has real coefficients, a nonzero real or imaginary part of a corresponding eigenfunction is again an eigenfunction. Normalize it in $L^2(\mu)$; orthogonality to the constants gives mean zero. On the centered subspace the spectrum lies in $[\lambda, \infty)$, so $\|H^{-1/2}\| \leq \lambda^{-1/2}$. Spectral calculus then gives $Hg = H^{1/2}h$ and $H^{1/2}g = h$ for centered $h \in \text{Dom}(H^{1/2})$. The first identity proves $g \in \text{Dom}(H)$ and the asserted norm equality; the second, together with the Dirichlet-form identity, proves the gradient equality. $\square$

The localization argument so far applies to smooth laws with positive lower and finite upper Hessian bounds. The next lemma approximates an arbitrary isotropic log-concave law by laws of this type without changing the dimension or isotropic normalization.

Lemma 3.11 (Regular isotropic approximation). *Every isotropic log-concave probability measure μ on $\mathbb{R}^n$ is the weak limit of regular isotropic laws $d\mu_j = e^{-W_j}dx$ satisfying $a_j I \preceq D^2 W_j \preceq b_j I$ for constants $0 < a_j \leq b_j < \infty$. The laws may be obtained by Gaussian convolution, a positive quadratic tilt, and an affine normalization whose translation tends to zero and whose linear part tends to I .*

Proof. Let $X \sim \mu$ and let Z be an independent standard Gaussian. For $\delta > 0$, write ν_δ for the law of $X + \sqrt{\delta}Z$. Its density e^{-V_δ} is positive and smooth. It is log-concave by preservation of log-concavity under convolution. Differentiating the Gaussian convolution under the integral gives

$$D^2 V_\delta(y) = \delta^{-1} I - \delta^{-2} \text{Cov}(X \mid X + \sqrt{\delta}Z = y).$$

Indeed, the gradient of the logarithm of the convolution density is $\delta^{-1}(\mathbb{E}[X \mid X + \sqrt{\delta}Z = y] - y)$, and the derivative of the conditional mean is the conditional covariance divided by δ . Gaussian damping justifies these differentiations locally uniformly in y . Convexity of V_δ and positivity of the conditional covariance therefore give $0 \leq D^2 V_\delta \preceq \delta^{-1} I$.

For $\epsilon > 0$, define

$$d\nu_{\delta,\epsilon}(y) := Z_{\delta,\epsilon}^{-1} e^{-\epsilon|y|^2/2} d\nu_\delta(y),$$

where $Z_{\delta,\epsilon} := \int e^{-\epsilon|y|^2/2} d\nu_\delta(y)$. The new potential $W_{\delta,\epsilon} := V_\delta + \epsilon|y|^2/2 + \log Z_{\delta,\epsilon}$ has Hessian between ϵI and $(\delta^{-1} + \epsilon)I$. For fixed δ , dominated convergence as $\epsilon \downarrow 0$ gives total-variation convergence to

ν_δ and convergence of its first two moments: apply it with weights 1, y_i , and $y_i y_k$. These weights are integrable since $X + \sqrt{\delta}Z$ has finite second moment. Choose $\delta_j \downarrow 0$ and then $\epsilon_j > 0$ so that the total-variation error and all first- and second-moment errors are at most $1/j$. Write $\nu_j := \nu_{\delta_j, \epsilon_j}$, and denote its mean by m_j and covariance by A_j . The coupling $X + \sqrt{\delta_j}Z \rightarrow X$ in L^2 implies $\nu_j \Rightarrow \mu$, $m_j \rightarrow 0$, and $A_j \rightarrow I$.

Each A_j is positive definite because ν_j has a positive density. Let μ_j be the law of $A_j^{-1/2}(Y - m_j)$ for $Y \sim \nu_j$. It is isotropic, and its smooth potential has Hessian $A_j^{1/2} D^2 W_{\delta_j, \epsilon_j}(m_j + A_j^{1/2} x) A_j^{1/2}$. Thus one may take $a_j := \epsilon_j \lambda_{\min}(A_j) > 0$ and $b_j := (\delta_j^{-1} + \epsilon_j) \lambda_{\max}(A_j) < \infty$. Since $A_j^{-1/2} \rightarrow I$ and $m_j \rightarrow 0$, these affine maps converge to the identity uniformly on compact sets. Tightness of ν_j then implies $\mu_j \Rightarrow \mu$, as asserted. $\square$

Approximation alone does not transfer an inequality to its full test class. The next lemma passes a uniform Poincaré bound through weak convergence and then removes compact-support and boundedness assumptions on the test function.

Lemma 3.12 (Stability of the Poincaré inequality). *Let $\mu_j \Rightarrow \mu$ be probability measures on $\mathbb{R}^n$, where μ is absolutely continuous with respect to Lebesgue measure. Suppose that a finite $K \geq 0$ satisfies $\text{Var}_{\mu_j}[q] \leq K \mathbb{E}_{\mu_j}[|\nabla q|^2]$ for every $q \in C_c^\infty(\mathbb{R}^n)$ and every j . Then every real locally Lipschitz q with $\mathbb{E}_\mu[|\nabla q|^2] < \infty$ belongs to $L^2(\mu)$ and satisfies*

$$\text{Var}_\mu[q] \leq K \mathbb{E}_\mu[|\nabla q|^2].$$

In particular, $C_P(\mu) \leq K$.

Proof. For a fixed $q \in C_c^\infty$, the functions q , q^2 , and $|\nabla q|^2$ are bounded and continuous. Weak convergence therefore passes the assumed inequality to μ with the same constant K .

Let now q be locally Lipschitz with finite Dirichlet energy, and set $q_M := \max\{-M, \min\{q, M\}\}$ for $M > 0$. The weak chain rule gives $|\nabla q_M| \leq |\nabla q|$ almost everywhere. With the cutoffs used in Lemma 3.9, the function $q_{M,R} := \chi_R q_M$ is compactly supported and globally Lipschitz. Convolve it with a smooth compactly supported approximate identity. The mollifications and their gradients converge Lebesgue-almost everywhere to $q_{M,R}$ and its gradient, respectively, and are uniformly bounded for fixed M, R , with supports in one compact set. Absolute continuity of μ and dominated convergence pass the compact-smooth inequality to $q_{M,R}$.

As $R \rightarrow \infty$, bounded convergence gives $q_{M,R} \rightarrow q_M$ in $L^2(\mu)$. The identity $\nabla q_{M,R} = \chi_R \nabla q_M + q_M \nabla \chi_R$ shows convergence of the gradients in $L^2(\mu)$: dominated convergence handles the first term, and the second has norm at most CM/R . Hence $\text{Var}_\mu[q_M] \leq K \mathbb{E}_\mu[|\nabla q_M|^2]$. For independent $X, X' \sim \mu$, Fatou's lemma now gives

$$\frac{1}{2} \mathbb{E}[(q(X) - q(X'))^2] \leq \liminf_{M \rightarrow \infty} \text{Var}_\mu[q_M] \leq K \mathbb{E}_\mu[|\nabla q|^2].$$

The first inequality uses pointwise convergence of the truncations and the two-copy formula for their variances. The second uses the inequality for q_M and $|\nabla q_M| \leq |\nabla q|$. The right side is finite, so Fubini gives a point y for which $\int (q(x) - q(y))^2 d\mu(x) < \infty$. Since $q(y)$ is finite, this implies $q \in L^2(\mu)$; the two-copy expression is then exactly $\text{Var}_\mu[q]$. $\square$

Proof of Theorem 3.1. Section 3.6 establishes $C_P(\nu) \leq C(\log(en))^{1/3}$ for every regular isotropic law ν , with one universal constant C . For an arbitrary isotropic log-concave μ , take the regular isotropic sequence from Lemma 3.11. An isotropic log-concave law is full-dimensional and hence has a density. Lemma 3.12, with $K := C(\log(en))^{1/3}$, therefore gives the same Poincaré bound for

μ and its full locally Lipschitz test class. Finally, Lemma 2.2, in the form of Eq. (7), yields $\psi_\mu \leq \sqrt{\pi C}(\log(en))^{1/6}$. Taking the supremum over isotropic log-concave μ proves the asserted bound for ψ_n . Only the scalar Poincaré inequality is passed to the limit; no continuity of eigenfunctions or inverse generators is required. For $n \geq 2$, the inequalities $\log n \leq \log(en) \leq (1 + 1/\log 2) \log n$ follow from $\log(en) = 1 + \log n$ and $\log n \geq \log 2$. Absorbing these fixed comparison factors into the universal constants gives exactly the logarithms stated in Theorem 1.1. $\square$

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